

- 3 (a) (i) By reference to the direction of propagation of energy, state how plane-polarised light differs from unpolarised light.

oscillations/vibrations (of all electric/magnetic vectors) of plane-polarised light in a single plane versus all possible planes for unpolarised light [B1]
 plane perpendicular/normal to direction of energy transfer / wave propagation [B1]
[2]

- (ii) Explain why sound waves cannot be polarised.

Sound waves are longitudinal waves [B1] (which cannot be polarised as vibrations are parallel to direction of energy transfer / wave propagation)
[1]

- (b) Light from a laser incident normally on an ideal polarising filter. The polarising filter is slowly rotated about a horizontal axis that is parallel to the incident light as shown in Fig. 3.1.

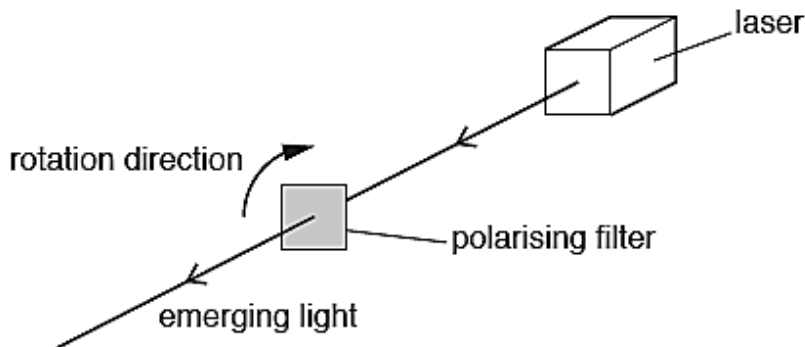


Fig. 3.1

It was observed that the intensity of the emerging light varies from I_0 to 0.

- (i) State what may be deduced from this observation.

Laser light is polarised [B1]
[1]

- (ii) On Fig. 3.2, sketch a graph to show how the intensity of the emergent light may vary with the angle θ which the filter is rotated. [2]

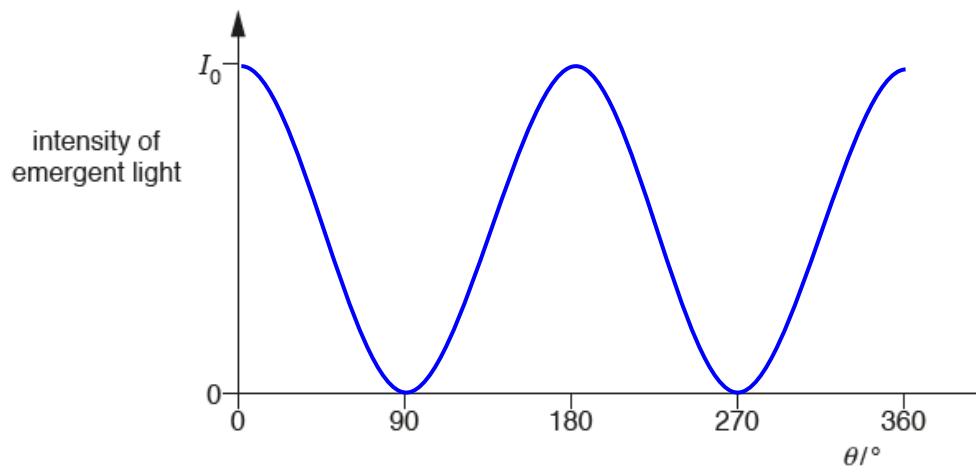


Fig. 3.2

sinusoidal square curve [B1]
2 complete cycles [B1]

- (c) Fig. 3.3 shows a beam of sunlight that is reflected from a surface and passes through a vertical polarising filter.

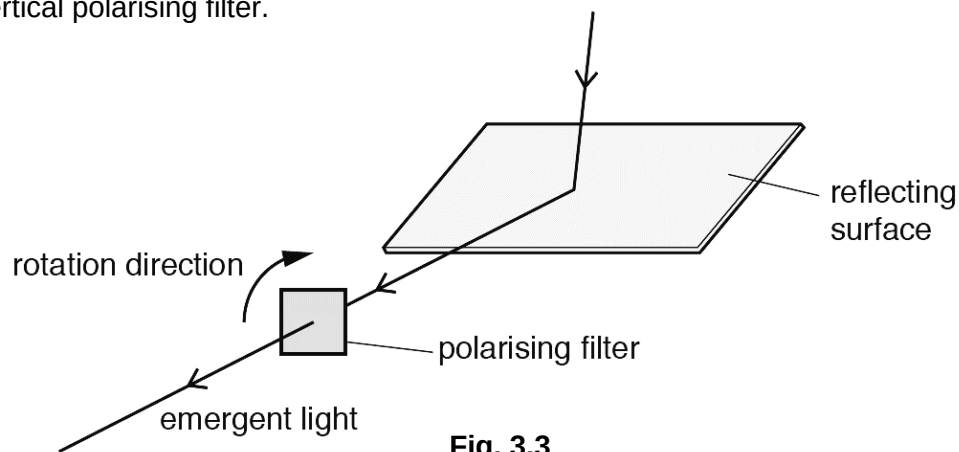


Fig. 3.3

The light that reflects from the surface is an unequal mixture of vertically polarised and horizontally polarised light.

When the polarising direction of the filter is vertical, the intensity of the emergent light is $0.262 I_0$. When the polarising direction of the filter is horizontal, the intensity is $0.850 I_0$.

Determine the intensity of the emergent light when the polarising direction of the filter is 30° from the vertical.

$$\begin{aligned} \text{Intensity} &= 0.262 I_0 \cos^2 30^\circ + 0.850 I_0 \sin^2 30^\circ \text{ [C1]} \\ &= 0.409 I_0 \text{ [A1]} \end{aligned}$$

intensity = I_0 [2]

[Total: 8]

- 4 (a) Explain what is meant by a *field of force*